CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the textbook exercises: 28 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 8.1

Q1 In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Find (i) $\sin A$, $\cos A$ (ii) $\sin C$, $\cos C$.

SOLUTION

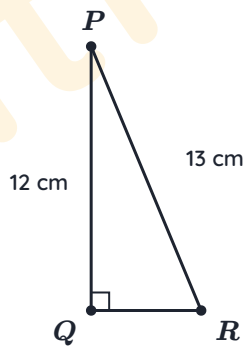
$AC = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$ cm.

(i) For angle A the opposite side is BC and the adjacent side is AB: $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$.

(ii) For angle C the opposite side is AB and the adjacent side is BC: $\sin C = \frac{24}{25}$, $\cos C = \frac{7}{25}$.

Answer: (i) $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25}$, $\cos C = \frac{7}{25}$

Q2 In $\triangle PQR$, right-angled at Q, $PQ = 12$ cm and $PR = 13$ cm (textbook Fig. 8.13). Find $\tan P - \cot R$.



SOLUTION

$QR = \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

$\tan P = \frac{QR}{PQ} = \frac{5}{12}$ and $\cot R = \frac{QR}{PQ} = \frac{5}{12}$.

Answer: $\tan P - \cot R = 0$

MathSetu

Q3 If $\sin A = \frac{3}{4}$, find $\cos A$ and $\tan A$.

SOLUTION

Take the opposite side as $3k$ and the hypotenuse as $4k$. The adjacent side is $\sqrt{16k^2 - 9k^2} = \sqrt{7}\,k$.

Answer: $\cos A = \frac{\sqrt{7}}{4}$ **and** $\tan A = \frac{3}{\sqrt{7}}$

Q4 Given $15 \cot A = 8$, find $\sin A$ and $\sec A$.

SOLUTION

$\cot A = \frac{8}{15}$, so take the adjacent side as $8k$ and the opposite side as $15k$. The hypotenuse is $\sqrt{64k^2 + 225k^2} = 17k$.

Answer: $\sin A = \frac{15}{17}$ **and** $\sec A = \frac{17}{8}$

Q5 Given $\sec \theta = \frac{13}{12}$, find all the other trigonometric ratios.

SOLUTION

Take the hypotenuse as $13k$ and the adjacent side as $12k$. The opposite side is $\sqrt{169k^2 - 144k^2} = 5k$.

Answer: $\sin \theta = \frac{5}{13}$, $\cos \theta = \frac{12}{13}$, $\tan \theta = \frac{5}{12}$, $\operatorname{cosec} \theta = \frac{13}{5}$, $\cot \theta = \frac{12}{5}$

Q6 $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$. Show that $\angle A = \angle B$.

SOLUTION

Take a triangle ABC right-angled at C, so that $\angle A$ and $\angle B$ are its acute angles. Then $\cos A = \frac{AC}{AB}$ and $\cos B = \frac{BC}{AB}$.

$\cos A = \cos B$ gives $AC = BC$.

Angles opposite equal sides of a triangle are equal, so $\angle B = \angle A$.

Hence $\angle A = \angle B$.

Q7 If $\cot \theta = \frac{7}{8}$, evaluate (i) $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$ (ii) $\cot^2 \theta$.

SOLUTION

(i) $\frac{1 - \sin^2 \theta}{1 - \cos^2 \theta} = \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta = \frac{49}{64}$.

(ii) $\cot^2 \theta = \left(\frac{7}{8}\right)^2 = \frac{49}{64}$.

Answer: (i) $\frac{49}{64}$ **(ii)** $\frac{49}{64}$

Q8 If $3\cot A = 4$, check whether $\dfrac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$.

SOLUTION

$\cot A = \dfrac{4}{3}$, so the adjacent side is $4k$, the opposite side is $3k$ and the hypotenuse is $5k$: $\tan A = \dfrac{3}{4}$, $\sin A = \dfrac{3}{5}$, $\cos A = \dfrac{4}{5}$.

LHS $= \dfrac{1-\frac{9}{16}}{1+\frac{9}{16}} = \dfrac{7}{25}$. RHS $= \dfrac{16}{25} - \dfrac{9}{25} = \dfrac{7}{25}$.

Answer: yes, both sides equal $\dfrac{7}{25}$

Q9 In $\triangle ABC$, right-angled at B, $\tan A = \dfrac{1}{\sqrt{3}}$. Find (i) $\sin A\cos C + \cos A \sin C$ (ii) $\cos A \cos C - \sin A \sin C$.

SOLUTION

Take $BC = k$ and $AB = \sqrt{3}\,k$, so $AC = \sqrt{k^2 + 3k^2} = 2k$.

$\sin A = \dfrac{1}{2}$, $\cos A = \dfrac{\sqrt{3}}{2}$, $\sin C = \dfrac{\sqrt{3}}{2}$, $\cos C = \dfrac{1}{2}$.

(i) $\dfrac{1}{2}\times\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\times\dfrac{\sqrt{3}}{2} = \dfrac{1}{4} + \dfrac{3}{4} = 1$. (ii) $\dfrac{\sqrt{3}}{2}\times\dfrac{1}{2} - \dfrac{1}{2}\times\dfrac{\sqrt{3}}{2} = 0$.

Answer: (i) 1 (ii) 0

Q10 In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Find $\sin P$, $\cos P$ and $\tan P$.

SOLUTION

Let $QR = x$ cm, so $PR = (25 - x)$ cm. By the Pythagoras theorem, $(25-x)^2 = 5^2 + x^2$.

$625 - 50x = 25$ gives $x = 12$. So $QR = 12$ cm and $PR = 13$ cm.

Answer: $\sin P = \dfrac{12}{13}$, $\cos P = \dfrac{5}{13}$, $\tan P = \dfrac{12}{5}$

Q11 State whether each is true or false, with a reason. (i) The value of $\tan A$ is always less than 1. (ii) $\sec A = \dfrac{12}{5}$ for some value of angle A. (iii) $\cos A$ is the abbreviation for the cosecant of angle A. (iv) $\cot A$ is the product of cot and A. (v) $\sin\theta = \dfrac{4}{3}$ for some angle θ.

SOLUTION

(i) False. For example, $\tan 60^\circ = \sqrt{3} > 1$.

(ii) True. $\sec A$ is the hypotenuse divided by the adjacent side, which can be any number from 1 upwards, and $\dfrac{12}{5} > 1$.

(iii) False. $\cos A$ stands for the cosine of A. The cosecant is written $\operatorname{cosec} A$.

(iv) False. $\cot A$ is one symbol, the cotangent of the angle A. "cot" alone has no meaning.

(v) False. $\sin\theta$ is the opposite side divided by the hypotenuse, and the hypotenuse is the longest side, so $\sin\theta$ cannot be more than 1.

Answer: (i) False (ii) True (iii) False (iv) False (v) False

2 Exercise 8.2

Q1 Evaluate: (i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ (ii) $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ (iii) $\dfrac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$ (iv) $\dfrac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$ (v) $\dfrac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.

SOLUTION

(i) $\dfrac{\sqrt{3}}{2} \times \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{3}{4} + \dfrac{1}{4} = 1.$

(ii) $2(1)^2 + \dfrac{3}{4} - \dfrac{3}{4} = 2.$

(iii) $\dfrac{\frac{1}{\sqrt{2}}}{\frac{2}{\sqrt{3}} + 2} = \dfrac{\sqrt{3}}{2\sqrt{2}\,(\sqrt{3}+1)}$. Multiply the top and bottom by $(\sqrt{3}-1)$: $\dfrac{\sqrt{3}\,(\sqrt{3}-1)}{4\sqrt{2}} = \dfrac{3-\sqrt{3}}{4\sqrt{2}} = \dfrac{3\sqrt{2}-\sqrt{6}}{8}$.

(iv) $\dfrac{\frac{1}{2} + 1 - \frac{2}{\sqrt{3}}}{\frac{2}{\sqrt{3}} + \frac{1}{2} + 1} = \dfrac{3\sqrt{3}-4}{3\sqrt{3}+4}$. Multiply the top and bottom by $(3\sqrt{3}-4)$: $\dfrac{(3\sqrt{3}-4)^2}{27-16} = \dfrac{43-24\sqrt{3}}{11}$.

(v) $\dfrac{5 \times \frac{1}{4} + 4 \times \frac{4}{3} - 1}{\frac{1}{4} + \frac{3}{4}} = \dfrac{15+64-12}{12} = \dfrac{67}{12}$.

Answer: (i) 1 (ii) 2 (iii) $\dfrac{3\sqrt{2}-\sqrt{6}}{8}$ (iv) $\dfrac{43-24\sqrt{3}}{11}$ (v) $\dfrac{67}{12}$

Q2 Choose the correct option and justify it. (i) $\dfrac{2\tan 30^\circ}{1+\tan^2 30^\circ}$ = (A) $\sin 60^\circ$ (B) $\cos 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$.
(ii) $\dfrac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ = (A) $\tan 90^\circ$ (B) 1 (C) $\sin 45^\circ$ (D) 0. (iii) $\sin 2A = 2\sin A$ is true when A = (A) 0° (B) 30° (C) 45° (D) 60°. (iv) $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ = (A) $\cos 60^\circ$ (B) $\sin 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$.

SOLUTION

(i) $\dfrac{2 \times \frac{1}{\sqrt{3}}}{1+\frac{1}{3}} = \dfrac{2}{\sqrt{3}} \times \dfrac{3}{4} = \dfrac{\sqrt{3}}{2} = \sin 60^\circ$. Option (A).

(ii) $\dfrac{1-1}{1+1} = 0$. Option (D).

(iii) For $A = 0^\circ$: $\sin 0^\circ = 0$ and $2\sin 0^\circ = 0$, so it holds. For the other options it fails, for example $\sin 60^\circ \neq 2\sin 30^\circ$. Option (A).

(iv) $\dfrac{2 \times \frac{1}{\sqrt{3}}}{1-\frac{1}{3}} = \dfrac{2}{\sqrt{3}} \times \dfrac{3}{2} = \sqrt{3} = \tan 60^\circ$. Option (C).

Answer: (i) A (ii) D (iii) A (iv) C

Q3 If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \dfrac{1}{\sqrt{3}}$, with $0^\circ < A+B \leq 90^\circ$ and $A > B$, find A and B.

SOLUTION

$\tan(A+B) = \sqrt{3} = \tan 60^\circ$ gives $A+B = 60^\circ$. $\tan(A-B) = \dfrac{1}{\sqrt{3}} = \tan 30^\circ$ gives $A-B = 30^\circ$.

Adding, $2A = 90^\circ$, so $A = 45^\circ$ and $B = 15^\circ$.

Answer: $A = 45^\circ$, $B = 15^\circ$

Q4 State whether each is true or false, with a reason. (i) $\sin(A+B) = \sin A + \sin B$. (ii) The value of $\sin\theta$ increases as θ increases. (iii) The value of $\cos\theta$ increases as θ increases. (iv) $\sin\theta = \cos\theta$ for all values of θ. (v) $\cot A$ is not defined for $A = 0°$.

SOLUTION

(i) False. With $A = 60°$, $B = 30°$: $\sin 90° = 1$ but $\sin 60° + \sin 30° = \dfrac{\sqrt{3}+1}{2} \neq 1$.

(ii) True for $0° \leq \theta \leq 90°$: the values go $0, \dfrac{1}{2}, \dfrac{1}{\sqrt{2}}, \dfrac{\sqrt{3}}{2}, 1$.

(iii) False. From $0°$ to $90°$ the values of $\cos\theta$ go from 1 down to 0.

(iv) False. They are equal only for $\theta = 45°$. For example $\sin 30° = \dfrac{1}{2}$ but $\cos 30° = \dfrac{\sqrt{3}}{2}$.

(v) True. $\cot 0° = \dfrac{\cos 0°}{\sin 0°} = \dfrac{1}{0}$, which is not defined.

Answer: (i) False (ii) True (iii) False (iv) False (v) True

3 Exercise 8.3

Q1 Express $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

SOLUTION

$\operatorname{cosec}^2 A = 1 + \cot^2 A$, so $\sin A = \dfrac{1}{\operatorname{cosec} A} = \dfrac{1}{\sqrt{1+\cot^2 A}}$.

$\tan A = \dfrac{1}{\cot A}$.

$\sec^2 A = 1 + \tan^2 A = 1 + \dfrac{1}{\cot^2 A}$, so $\sec A = \dfrac{\sqrt{1+\cot^2 A}}{\cot A}$.

Answer: $\sin A = \dfrac{1}{\sqrt{1+\cot^2 A}}$, $\sec A = \dfrac{\sqrt{1+\cot^2 A}}{\cot A}$, $\tan A = \dfrac{1}{\cot A}$

Q2 Write all the other trigonometric ratios of $\angle A$ in terms of $\sec A$.

SOLUTION

$\cos A = \dfrac{1}{\sec A}$.

$\tan^2 A = \sec^2 A - 1$, so $\tan A = \sqrt{\sec^2 A - 1}$ and $\cot A = \dfrac{1}{\sqrt{\sec^2 A - 1}}$.

$\sin A = \tan A \times \cos A = \dfrac{\sqrt{\sec^2 A - 1}}{\sec A}$ and $\operatorname{cosec} A = \dfrac{\sec A}{\sqrt{\sec^2 A - 1}}$.

Answer: $\cos A = \dfrac{1}{\sec A}$, $\sin A = \dfrac{\sqrt{\sec^2 A - 1}}{\sec A}$, $\tan A = \sqrt{\sec^2 A - 1}$, $\cot A = \dfrac{1}{\sqrt{\sec^2 A - 1}}$, $\operatorname{cosec} A = \dfrac{\sec A}{\sqrt{\sec^2 A - 1}}$

MathSetu

Q3 Choose the correct option and justify it. (i) $9\sec^2 A - 9\tan^2 A =$ (A) 1 (B) 9 (C) 8 (D) 0. (ii) $(1+\tan\theta+\sec\theta)(1+\cot\theta-\operatorname{cosec}\theta) =$ (A) 0 (B) 1 (C) 2 (D) -1. (iii) $(\sec A+\tan A)(1-\sin A) =$ (A) $\sec A$ (B) $\sin A$ (C) $\operatorname{cosec} A$ (D) $\cos A$. (iv) $\dfrac{1+\tan^2 A}{1+\cot^2 A} =$ (A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$.

SOLUTION

(i) $9(\sec^2 A - \tan^2 A) = 9 \times 1 = 9$. Option (B).

(ii) $\dfrac{\cos\theta+\sin\theta+1}{\cos\theta} \times \dfrac{\sin\theta+\cos\theta-1}{\sin\theta} = \dfrac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta} = \dfrac{2\sin\theta\cos\theta}{\sin\theta\cos\theta} = 2$. Option (C).

(iii) $\dfrac{1+\sin A}{\cos A} \times (1-\sin A) = \dfrac{1-\sin^2 A}{\cos A} = \cos A$. Option (D).

(iv) $\dfrac{\sec^2 A}{\operatorname{cosec}^2 A} = \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$. Option (D).

Answer: (i) B (ii) C (iii) D (iv) D

Q4 (i) Prove that $(\operatorname{cosec}\theta - \cot\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$.

SOLUTION

$$\text{LHS} = \left(\frac{1-\cos\theta}{\sin\theta}\right)^2 = \frac{(1-\cos\theta)^2}{1-\cos^2\theta} = \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)}$$

$$= \frac{1-\cos\theta}{1+\cos\theta} = \text{RHS}$$

Q4 (ii) Prove that $\dfrac{\cos A}{1+\sin A} + \dfrac{1+\sin A}{\cos A} = 2\sec A$.

SOLUTION

$$\text{LHS} = \frac{\cos^2 A + (1+\sin A)^2}{(1+\sin A)\cos A} = \frac{\cos^2 A + \sin^2 A + 1 + 2\sin A}{(1+\sin A)\cos A} = \frac{2(1+\sin A)}{(1+\sin A)\cos A}$$

$$= \frac{2}{\cos A} = 2\sec A = \text{RHS}$$

Q4 (iii) Prove that $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\operatorname{cosec}\theta$.

SOLUTION

Write everything with $\sin\theta$ and $\cos\theta$: $\text{LHS} = \dfrac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \dfrac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)} = \dfrac{\sin^3\theta - \cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$

Using $a^3-b^3 = (a-b)(a^2+ab+b^2)$: $= \dfrac{\sin^2\theta+\sin\theta\cos\theta+\cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta}$

$= \sec\theta\operatorname{cosec}\theta + 1 = \text{RHS}$

Q4 (iv) Prove that $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$.

SOLUTION

LHS $= \frac{1}{\sec A} + 1 = \cos A + 1$.

RHS $= \frac{1-\cos^2 A}{1-\cos A} = \frac{(1-\cos A)(1+\cos A)}{1-\cos A} = 1+\cos A$.

LHS = RHS

Q4 (v) Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using $\operatorname{cosec}^2 A = 1 + \cot^2 A$.

SOLUTION

Divide the numerator and the denominator by $\sin A$: LHS $= \frac{\cot A - 1 + \operatorname{cosec} A}{\cot A + 1 - \operatorname{cosec} A}$.

In the numerator replace 1 by $\operatorname{cosec}^2 A - \cot^2 A$: $(\cot A + \operatorname{cosec} A) - (\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A) = (\cot A + \operatorname{cosec} A)(1 - \operatorname{cosec} A + \cot A)$.

So LHS $= \frac{(\cot A + \operatorname{cosec} A)(\cot A + 1 - \operatorname{cosec} A)}{\cot A + 1 - \operatorname{cosec} A} = \operatorname{cosec} A + \cot A =$ **RHS**

Q4 (vi) Prove that $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$.

SOLUTION

Multiply the numerator and the denominator inside the root by $(1+\sin A)$: LHS $= \sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}} = \frac{1+\sin A}{\cos A}$

$= \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \tan A =$ **RHS**

Q4 (vii) Prove that $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$.

SOLUTION

LHS $= \frac{\sin\theta\,(1-2\sin^2\theta)}{\cos\theta\,(2\cos^2\theta - 1)}$.

Using $1 = \sin^2\theta + \cos^2\theta$: $1 - 2\sin^2\theta = \cos^2\theta - \sin^2\theta$ and $2\cos^2\theta - 1 = \cos^2\theta - \sin^2\theta$.

So LHS $= \frac{\sin\theta}{\cos\theta} = \tan\theta =$ **RHS**

Q4 (viii) Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

SOLUTION

LHS $= \sin^2 A + \operatorname{cosec}^2 A + 2 + \cos^2 A + \sec^2 A + 2$, since $\sin A \times \operatorname{cosec} A = 1$ and $\cos A \times \sec A = 1$.

$= (\sin^2 A + \cos^2 A) + 4 + (1 + \cot^2 A) + (1 + \tan^2 A)$

$= 7 + \tan^2 A + \cot^2 A =$ **RHS**

Q4 (ix) Prove that $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$.

SOLUTION

$$\text{LHS} = \frac{1-\sin^2 A}{\sin A} \times \frac{1-\cos^2 A}{\cos A} = \frac{\cos^2 A \sin^2 A}{\sin A \cos A} = \sin A \cos A.$$

$$\text{RHS} = \frac{1}{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}} = \frac{\sin A \cos A}{\sin^2 A + \cos^2 A} = \sin A \cos A.$$

LHS = RHS

Q4 (x) Prove that $\dfrac{1+\tan^2 A}{1+\cot^2 A} = \left(\dfrac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$.

SOLUTION

First expression: $\dfrac{\sec^2 A}{\operatorname{cosec}^2 A} = \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$.

Second expression: $1 - \cot A = \dfrac{\tan A - 1}{\tan A}$, so $\dfrac{1-\tan A}{1-\cot A} = \dfrac{(1-\tan A)\tan A}{\tan A - 1} = -\tan A$, and its square is $\tan^2 A$.

Both expressions equal $\tan^2 A$.

The questions are restated in MathSetu's own words and the figures are redrawn; refer to the textbook for the original wording. The solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by NCERT, CBSE or any publisher.